EE 474 Introduction to Controls
Lecture Notes

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1 Introduction

1.1 What is "Controls"

Regulation of equipment or a process to achieve an objective.

1.2 Two simple examples:

1.2.1 Gas Stove

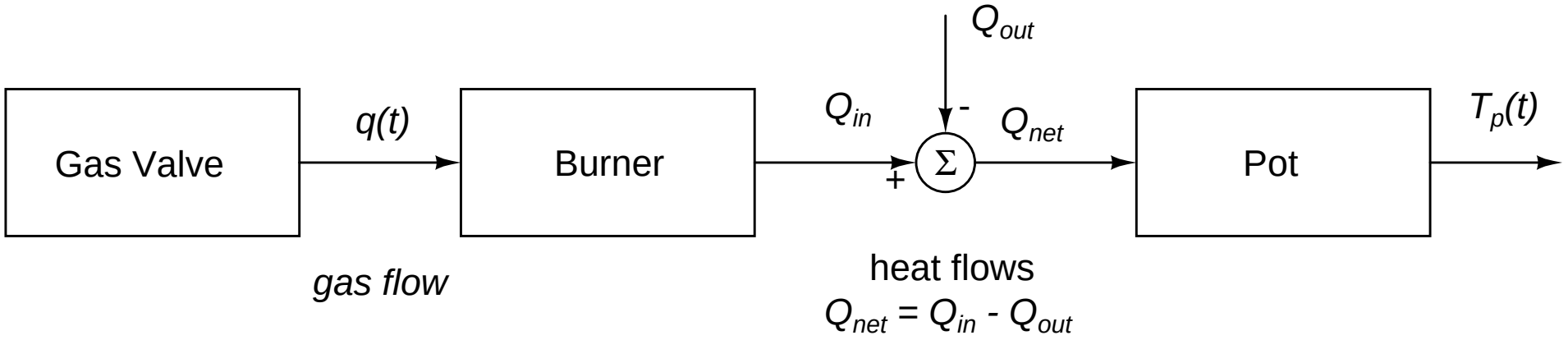


Figure 1: A typical gas stove is an example of ***open-loop control***.

- This is an example of an *Open-Loop* system

Components of a Block Diagram:

- Blocks are elements:
  - Rectangular blocks do processing (possible dynamics)
  - Circles are summing junctions
- Lines are signals.
- Both have physical units.

1.2.2 Room Temperature Control

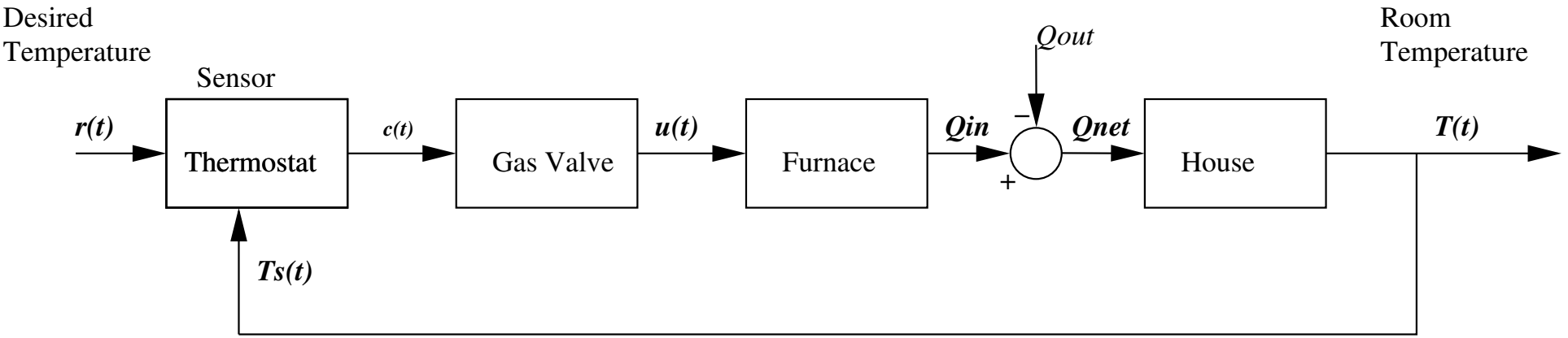


Figure 2: Room temperature, ***closed-loop control***.

Signal	Units	Description
$r(t)$	$[^0C]$	Reference Temperature
$c(t)$	[volts]	Command
$u(t)$	$[\mathrm{m}^3/\mathrm{hr}]$	Gas Flow
$Q_{in}(t)$	$[KW]$	Furnace Heat
$T(t)$	$[^0C]$	Room Temperature

Parameter	Units	Description
K_t	$\left[\frac{volts}{^oC}\right]$	Thermostat Coefficient
K_v	$\left[\frac{m^3/hr}{volt}\right]$	Gas Valve Coefficient
K_f	$\left[\frac{\mathrm{KW}}{\mathrm{m}^3/\mathrm{hr}}\right]$	Furnace Coefficient
C	$\left[\frac{^oC}{Joule}\right]$	Heat Capacitance of House

Table 1: Signals and Parameters of system in figure 2.

- Notice difference between signals and parameters
  - Signals:
    - A function of time (volts, $v(t)$, temperature, $T(t)$, etc.);
    - Describe what is happening;
    - Not known to the controls engineer in advance (the controls engineer must engineer the system for any possible signal).
  - Parameters:
    - Generally constants (else see adaptive control);
    - Describe component;
    - The controls engineer determines the structure and parameters of the controller.

2 Open-loop versus Closed-loop

- *Open-Loop* Control
  - Gas stove:

    Heat flux is what it is, independent of what happens.
- *Closed-Loop* control
  - Room-temperature Control:

    Heat flux responds to room temperature.

2.1 Examples of open- and closed-loop control

Car engine, fuel/air mixture

- Proper combustion in the cylinder depends on having the right mixture of fuel and air

Open Loop: Carburetor

- Carburetors were pretty complicated fluidic devices
  - They responded to air flow, air pressure and engine temperature
  - But they did not correct errors (over-lean or over-rich fuel/air mixture)

Closed Loop: Computer-controlled fuel injection

- Uses an exhaust O_2 sensor
- Continuously monitors fuel air mixture, and adjusts
  - Improves fuel economy and performance
  - Dramatically improves cold-engine starting
  - 30 x reduction in HC, 10 x reduction of other pollutants

3 Components of a closed-loop feedback control system

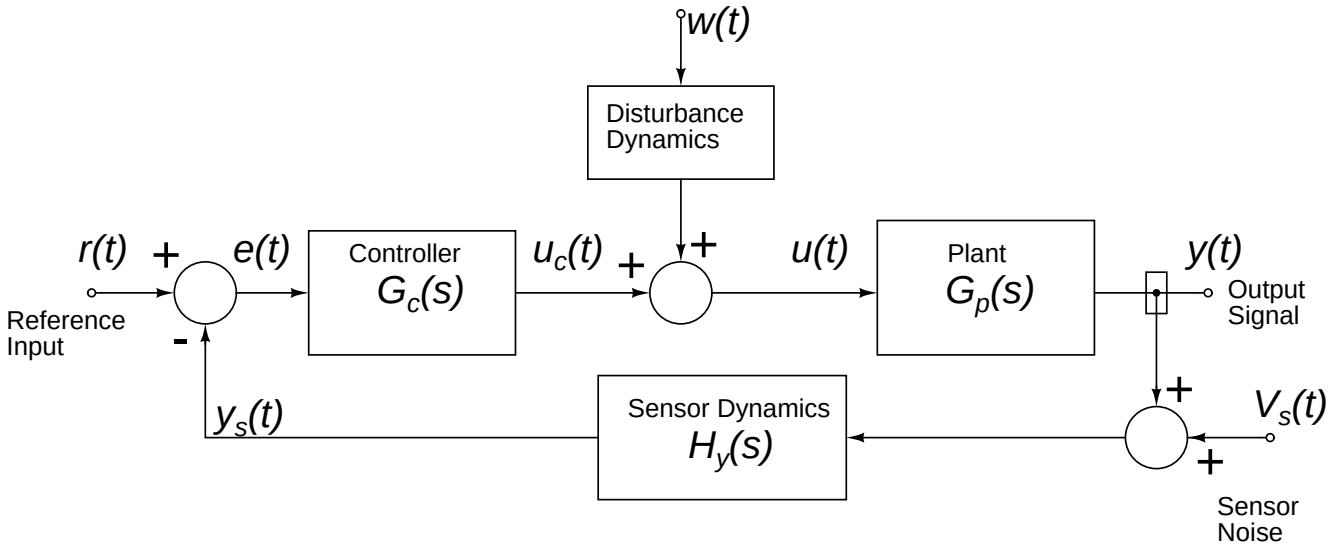


Figure 3: Functional block diagram of a closed-loop system.

Term	Meaning	Symbol
Process or Plant	The system whose output is to be controlled. (Such as Car speed).	$G_p(s)$
Controller	Element that generates control signals, designed by the Controls Engineer.	$G_c(s)$
Control signal / Control variable	Input signal to the plant, used to control it. Output from the controller. (Such as throttle angle).	$u(t)$
Controlled variable	Output of the plant which is controlled (such as car speed, note: car heading is another output, but is not controlled by cruise control.)	$y(t)$
Actuator	Device, such as a motor or heating element, which converts the control signal to action.	
Sensor	Detects controlled variable or other signals.	
Sensed signal	Output of the sensor.	$y_s(t)$
Reference signal	Desired value of controlled variable.	$r(t)$
Error signal	Difference between reference and measured, $e(t) = r(t) - y_s(t)$	$e(t)$
Disturbance signal	A signal entering the system separate from the control signal, disturbing the output	$w(t)$, $d(t)$

3.1 Example, based on automobile control

3.1.1 Open loop (Automobile with a fixed throttle)

Model:

$$y(t) = 10\,(u_c(t) + 0.5\,w(t)) \quad (1)$$

$$u_c(t) = G_c\,r(t) \quad (2)$$

List Signals and Parameters:

Signal	Units	Description
$y(t)$	[MPH]	Car speed
$u(t)$	[deg]	Throttle angle
$w(t)$	[%]	Road grade
$r(t)$	[MPH]	Reference speed

Table 2: Signals. Note: road grade (steepness) is measured in % rise over run.

Parameter	Units	Description
10.0	[MPH/deg]	Speed per throttle
0.5	[deg/%]	Impact of road grade
G_c	Deg/MPH	Controller gain

Table 3: Parameters for open-loop speed control.

Automobile open-loop control: functional block diagram

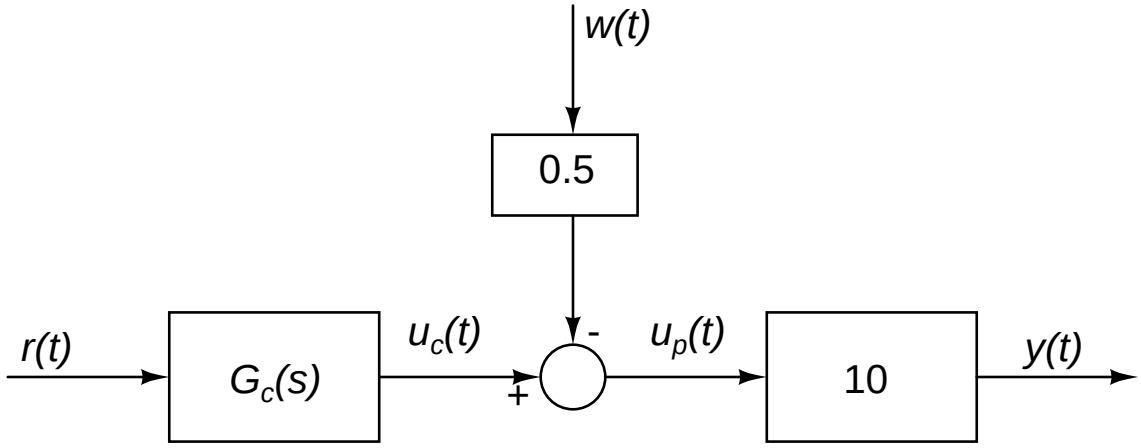


Figure 4: Functional block diagram automobile cruise control.

- The Block diagram is the Controls Engineers graphical language for mathematical relationships. The block diagram graphically shows the model relations:

$$y(t) = 10\,u_p(t) = 10\,(u_c(t) - 0.5\,w(t)) \tag{3}$$

$$u_c(t) = G_c\,r(t) \tag{4}$$

- In addition to relations (equations), block diagrams show causality by the arrows.

- Note units,

$$\begin{aligned} y(t) &= 10\left(u_c(t) - 0.5\,w(t)\right) \\ [MPH] &= \left[\tfrac{MPH}{deg}\right]\left([deg]\text{-}\left[\tfrac{\text{deg}}{\%}\right]\ [\%]\right) \\ u_c(t) &= G_c\,r(t) \\ [deg] &= \left[\tfrac{deg}{MPH}\right]\ [MPH] \end{aligned}$$

  - Both the terms in modeling equations and the elements and links in block diagrams have physical units.

Analysis for Open-Loop Case

Substituting $u(t)$ into the equation giving $y(t)$:

$$y(t) = 10.0\,(G_c\,r(t) - 0.5\,w(t)) = 10.0\,G_c\,r(t) - 5.0\,w(t) \tag{5}$$

For $y(t) = r(t)$ (when $w(t) = 0$) **choose** $G_c = 0.1$

$$y(t) = 1.0\,r(t) - 5.0\,w(t) \tag{6}$$

We can define the error:

$$e(t) = r(t) - y(t) \qquad [MPH] \tag{7}$$

Calculating error for $r(t) = 55$ [MPH]

Note: choosing $G_c = 0.1$ is a step of controller design.

Grade $w(t)$	$y(t)$	$e(t)$	% Error
0%	55.0	0.0	0.0%
1%	50.0	5.0	9.1%
2%	45.0	10.0	18.2%

Table 4: Example values, open loop, $r(t) = 55$ [MPH].

3.1.2 Closed Loop (Automobile cruise control)

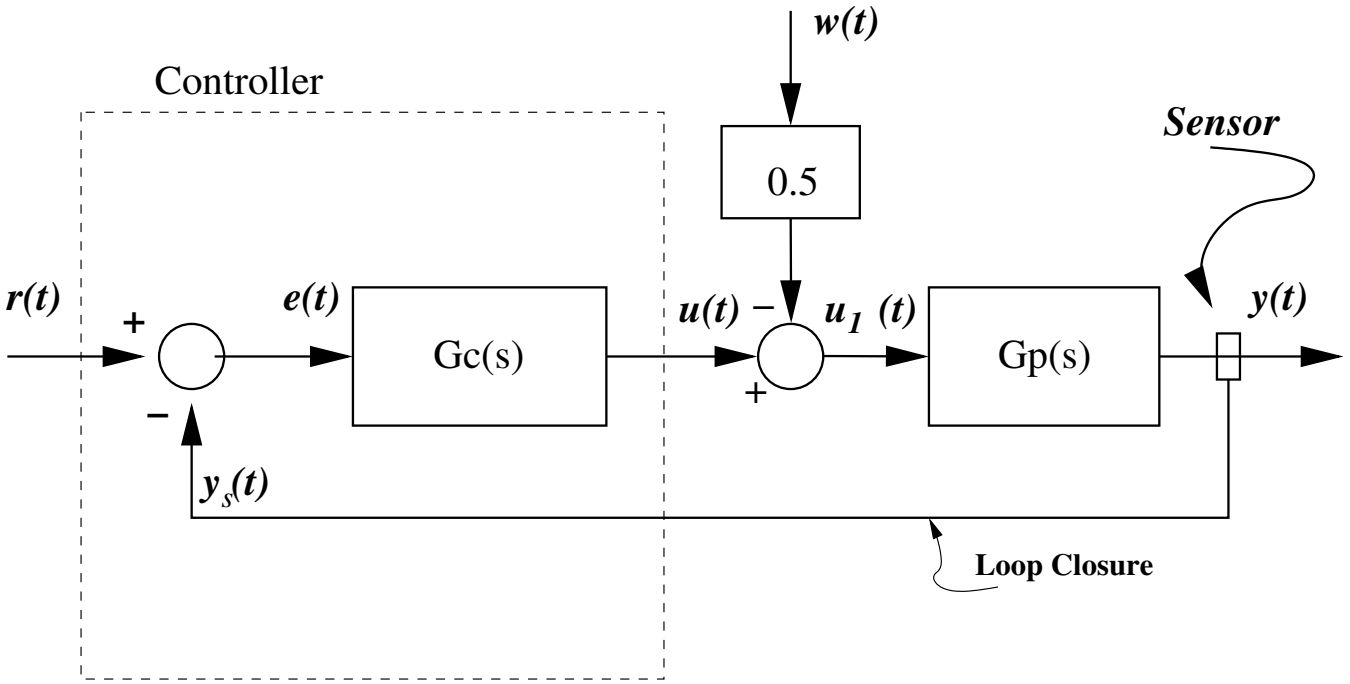


Figure 5: Functional block diagram.

The Controller may include several components, such as

- Reference command generator, $r(t)$
- Comparator, or summing junction (generates error signal),
- Amplifier,
- Actuator (e.g., throttle).
- Sensor(with feedback to the comparator)

Choose G_c

- A constant controller parameter, such as G_c, is called ***proportional control***.
- This is controller design, you will do this in Lab 1 !
- Choosing:

$$G_c = 100.0 \quad \text{Gives :} \quad u(t) = 100.0\,e(t) = 100.0\,(r(t) - y_s(t)) \tag{8}$$

Analysis for Closed-Loop Case

Substituting $u(t)$ into the equation giving $y(t)$:

$$y(t) = 10\,u(t) - 5\,w(t) = 10\,(100\,(r(t) - y(t))) - 5\,w(t) \tag{9}$$

Solving for $y(t)$ given $r(t)$:

$$\begin{aligned} y(t) &= 1000\,r(t) - 1000\,y(t) - 5\,w(t) \\ 1001\,y(t) &= 1000\,r(t) - 5\,w(t) \end{aligned}$$

Solving for system-output $y(t)$:

$$y(t) = \frac{1000}{1001}\,r(t) - \frac{5}{1001}\,w(t) = 0.999\,r(t) - 0.005\,w(t) \tag{10}$$

Calculating error for $r(t) = 55$ [MPH]

Grade $w(t)$	$y(t)$	$e(t)$	% Error	(Recall figures for open loop)	
0%	54.945	0.055	0.10%	55.0	0.0%
1%	54.940	0.060	0.11%	50.0	9.1%
2%	54.935	0.065	0.12%	45.0	18.2%

Table 5: Example values, closed loop, $r(t) = 55$ [MPH].

Observe:

1. Effect of disturbance is significantly reduced by feedback control
2. For $w = 0$, error $\neq 0$

Design Trade-off:

- $G_c = 100$ (a large value, gives a loop gain of 1000) $\Longrightarrow$ small errors
- But large control gains $\Longrightarrow$ **instability**

3.2 Controller Performance Measures

Aspects of basic controller performance break into four categories:

- Speed
- Accuracy
- Stability (damping)
- Robustness

Generally, the controls designer wants to achieve a fast, accurate, well damped and robust closed-loop system. And generally, controls design is a matter of making trade-offs between speed, accuracy, damping and robustness.

$\Longrightarrow$ The art of Controls Engineering is to make this trade-off to best advantage.

4 Semester plan for EE 474

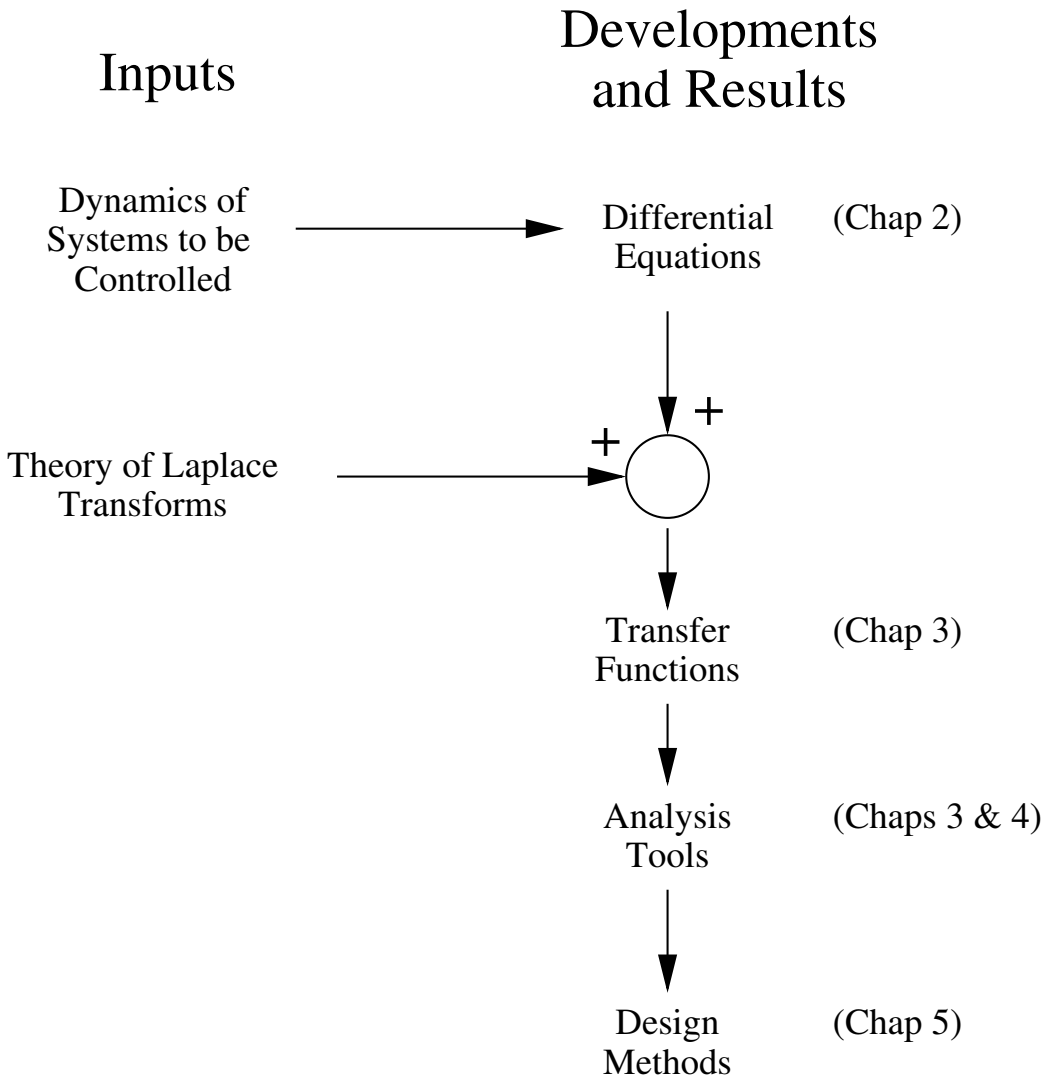


Figure 6: Road map to EE 474.